

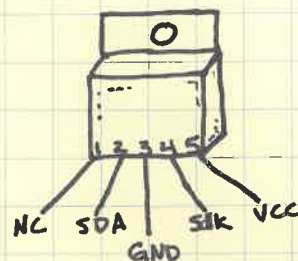
TC74 temperature Sensor

TC74_Temp_Sensor (code Name in Arduino)

```
#include <Wire.h>
byte tc74 = 0x4D;

void setup()
{
  Serial.begin(9600);
  Wire.begin(); //Start I2C ops
  delay(100);
}
```

```
void loop()
{
  Wire.beginTransmission(tc74);
  Wire.write(0x01); //tell it what register to write to
  Wire.write(0x00); //send the command for "normal" mode
  Wire.endTransmission(); //stop condition
  delay(100);
  Wire.beginTransmission(tc74); //start condition
  Wire.write(0x00); //tell it what register we are reading
  Wire.endTransmission(); //then have to stop
  Wire.requestFrom(tc74, 1); //repeated start reading 1 byte
  int tempC = Wire.read(); //read the data register
  Wire.endTransmission();
  Serial.println(tempC);
  delay(90000);
}
```



1. No Connection
2. SDA
3. Ground
4. SCL
5. VCC (Power)

I2C Address of TC74 $\Rightarrow$ 0x4D

Address Finder Code For I2C

```
#include <Wire.h> //use wire.h library
void setup()
{
  Serial.begin(9600);
  Wire.begin();
  for(int add = 0; add < 128; add++) //loop through all possible addresses
  {
    Wire.beginTransmission(add); //start condition
    if(Wire.endTransmission() == 0) //look for ACK for I2C device
    {
      Serial.print("0x");
      Serial.println(add, HEX);
    }
  }
}

void loop()
{
}
```

How I2C works $\rightarrow$ connects Master



to Slave device via 2 Lines
Each Requires a Pull-Up Resistor
holding Inactive Logic High
Between Communications
Each Slave has 7 Bit Address
That Transmits to Master at
Start of Each Transmission

Data To Slave

1. Master Sets a Start Condition
SDA Low SCL High

2. Master Follows Start Condition
Clocking 7-bit Slave Address
on SDA Line XXXX XXX 0 indicates
write operation is underway

3. Target Slave Responds w/ ACK $\Rightarrow$ 0
on SDA

4. Master Sends 8-bit register Address to Slave Master

5. Slave Acknowledges Reg Address Receipt

6. Master Sends Data Byte to Slave byte, issues NACK,

7. Slave Responds w/ ACK on SDA

8. Master issues Stop Condition (High-High)

Data From Slave

- 1-5 same

6. Start Condition Sent to Master
"Repeated Start"

7. Master transmits XXXX XXX 1 Bit

8. Slave Ack Response

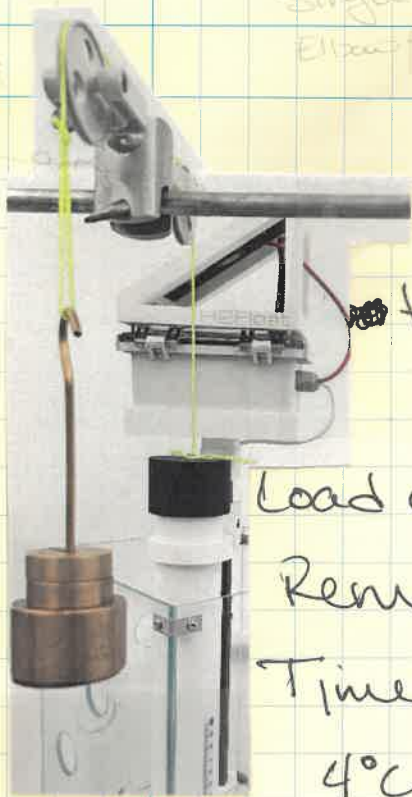
9. Slave byte transmits to

10. Master gets Data

- Stop Condition

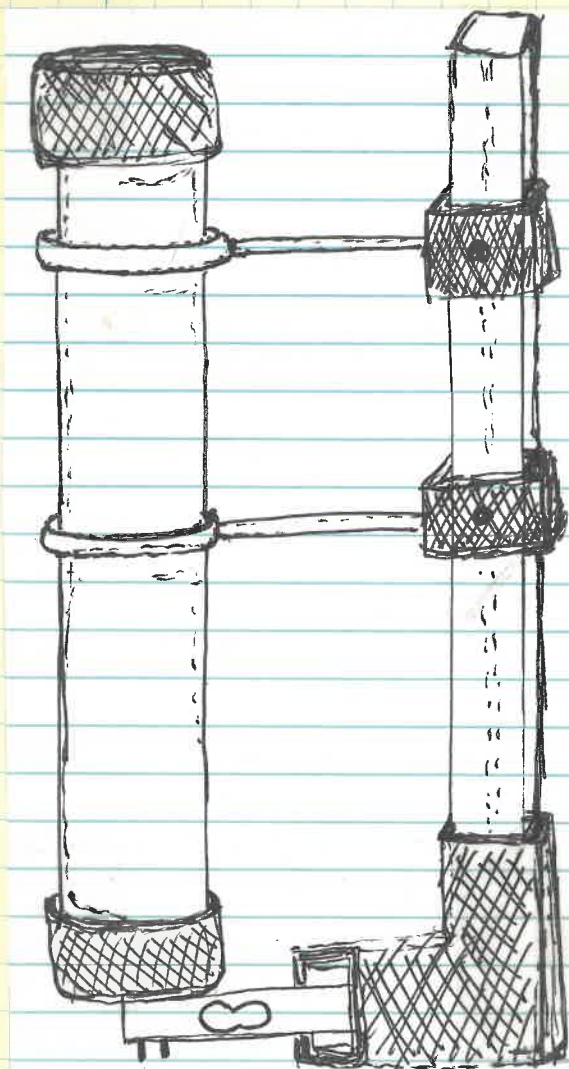
Single Load Cell Design including Support/measurements
 Elliptical Top/Bot Cap Drawing New Base Design
 Roll measurements

Considerations
 for sending to field
 Rustingable Parts



Solution
 to Temperature Issues:

Single Load Cell Design With
 Load cell mounted at Bottom of pipe to
 Remain Submerged in Water at all
 Times. The Temperature fluctuation is between
 4°C (40°F) and 27°C (80°F) Annually.



* New Design has two Pipe Supports
 that allows the Pipe to move
 Freely without tipping

Device Calibration

7/18/24 Calibration Air 22°C Water 21.5°C

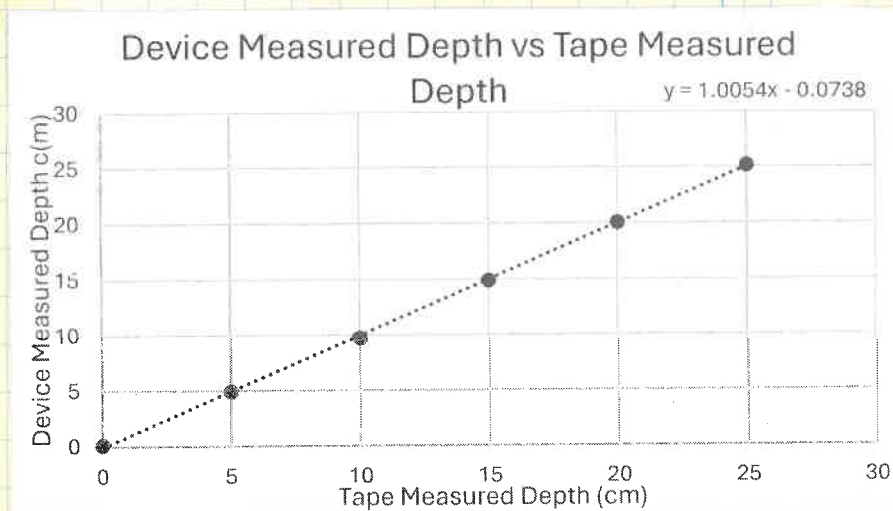
Tare	750g	Mass	Force	Depth
9090598	8546957	0.001	0.006	0
9090021	8546863	0.001	0.013	0.001
9088967	8546644	-0.002	-0.017	-0.001
9090065	8546078	0	-0.001	0
9090281	8546551	0.002	0.02	0.001
9088752	8545180	-0.001	-0.007	0
9089029	8545421	0	-0.002	0
9089126	8544932	0	0	0
9089113	8543455	0	-0.001	0
9089295	8543934	0	0.001	0
9088423	8544359	0	0.001	0
9088347	8544446	0	-0.002	0
9088495	8543421	0	0.002	0
9088163	8542762	-0.001	-0.007	0
9088802	8543827	0.001	0.013	0.001
9089165	8544989	7E-05	0.0013	1E-04

Checking accuracy of Calibration on Single Load Cell Design.

Calibration Values Used Air 22°C : Water 21.5°C

Tare	Weighted Tare
9089165	8544989

Tape Measured Depth	Device Measured Depth
0	0.13
5	4.92
10	9.75
15	14.89
20	20.07
25	25.20

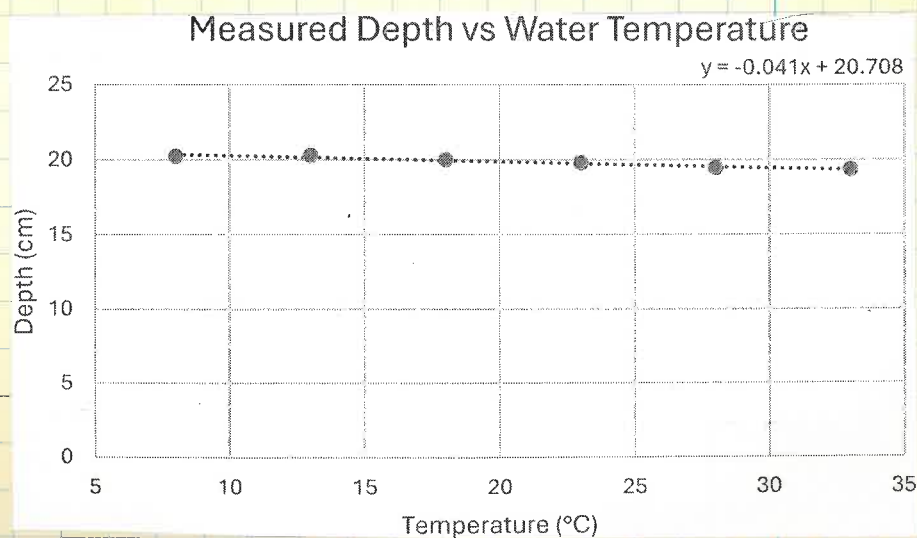


Testing Temperature Dependence

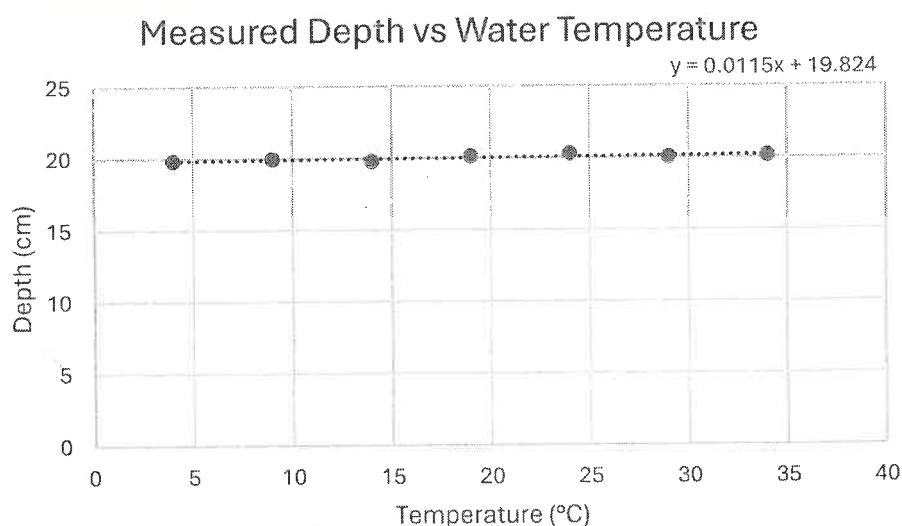
20cm Depth by Tape Measure		
Thermometer H ₂ O Temp °C	TC74 H ₂ O Temp °C	Depth
8	10	20.25
13	15	20.31
18	21	20.01
23	26	19.81
28	30	19.47
33	36	19.36

Over a 25°C Range
of Temperatures
The Variation in depth was
within a 1cm Range

The most frequent Water
Temperature Range of
15°C Variation up to
21°C the dependence
on temperature is
negligible and within
acceptable and expected
depth Variation of 1/2 cm.



20cm Depth by Tape Measure		
Thermometer H ₂ O Temp °C	TC74 H ₂ O Temp °C	Depth
4	7	19.84
9	13	19.98
14	17	19.80
19	23	20.16
24	28	20.31
29	32	20.05
34	37	20.16

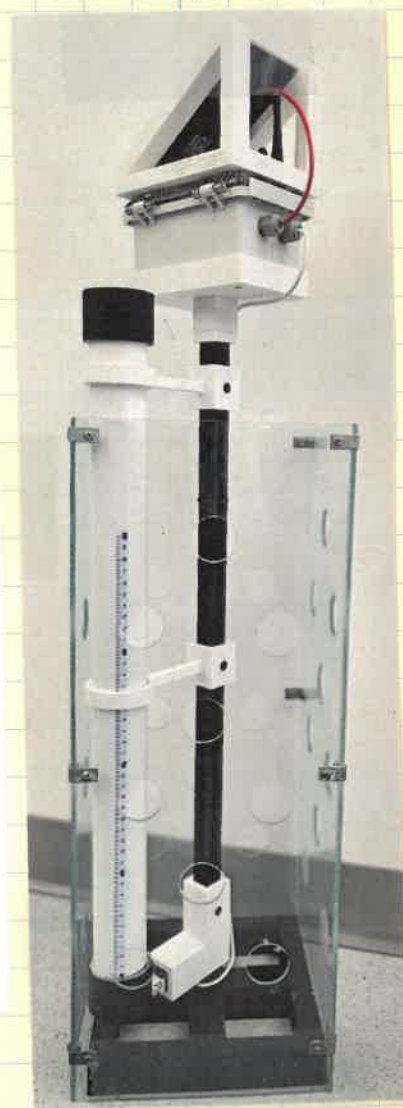


Over a larger Temperature
Range Variations the
Temperature dependence

decreased more
Proving the Concern
for Temperature
dependence is
no more with
The Single Load
Cell design.

```
#include <Q2HX711.h>           //Calls Hx711 Library
#include <SPI.h>                 //Calls SPI Library
#include <LoRa.h>               //Calls LoRa Library
#include <ArduinoLowPower.h>    //Calls Low Power Library
Q2HX711 HX711(2,3);           //Identifies HX711 pins (2B,3Y)
long Tare = 9089165;           //Averaged tare value for calibration
long WeightedTare = 8544989;   //Averaged Weighted tare value for calibration
float Battery;                 //Battery Variable for sending data over LoRa

void setup()
{
}
```




```

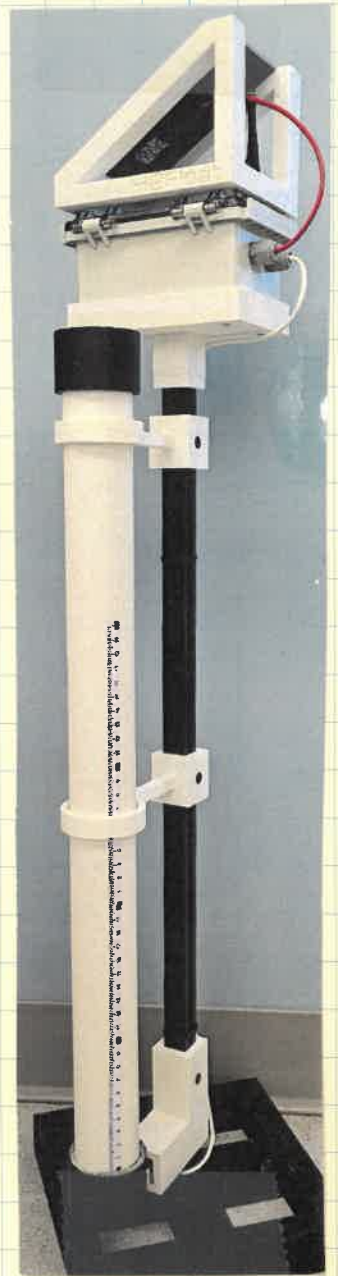
void loop()
{
  Serial.begin(9600);
  delay(3000);
  if(!LoRa.begin(915E6))           //915 MHz
  {
    Serial.println("Starting LoRa failed");
    while(1);                     //If LoRa fails, it jams up
  }
  Battery = measureBV();
  Serial.println("LoRa Started");
  long Reading = HX711.read();      //Load cell reading instance
  float Mass = 0.75*(Reading-Tare)/(WeightedTare-Tare); // Find Mass
  float Force = Mass*9.8;          // Find Force
  float Depth = (Force/18.5)*100.0; //Water level change (cm)
  Serial.print(Depth);             // Print to verify packet Data
  Serial.print("\t");
  Serial.println(Battery);
  LoRa.beginPacket();              // Creates packet to send
  LoRa.print('H');                 // Header Specifies H2Float Data
  LoRa.print('D');                 //Header identifies Depth Data
  LoRa.print(Depth);               // Data sent in packet
  LoRa.endPacket();                // Closes packet
  delay(1000);
  LoRa.beginPacket();              // Creates packet to send
  LoRa.print('H');                 // Header Specifies H2Float Data
  LoRa.print('B');                 // Header identifies Battery Voltage Data
  LoRa.print(Battery);             // Data sent in packet
  LoRa.endPacket();
  delay(1000);

  //Shut down LoRa
  digitalWrite(LORA_RESET,LOW);    //Take it low to conserve Power
  digitalWrite(LORA_BOOT0,LOW);
  pinMode(LORA_RESET,INPUT);
  pinMode(LORA_DEFAULT_SS_PIN,INPUT); //Slave Select
  pinMode(LORA_IRQ_DUMB,INPUT);    //Interrupt
  pinMode(LORA_BOOT0,INPUT);
  Serial.println("LoRa LOW");

  //Sleep MKR1310
  Serial.println("Sleep the Board...");
  Serial.end();
  LowPower.sleep(6000);            // How long Board sleeps
  }                                //Can't reupload the program while sleeping
                                  //Must double click reset button

float measureBV()                  //Creates Battery Voltage variable
{
  float Volts = 2.0*3.3*analogRead(A6)/1023.0; //Calculates current battery voltage
  return Volts;
}

```



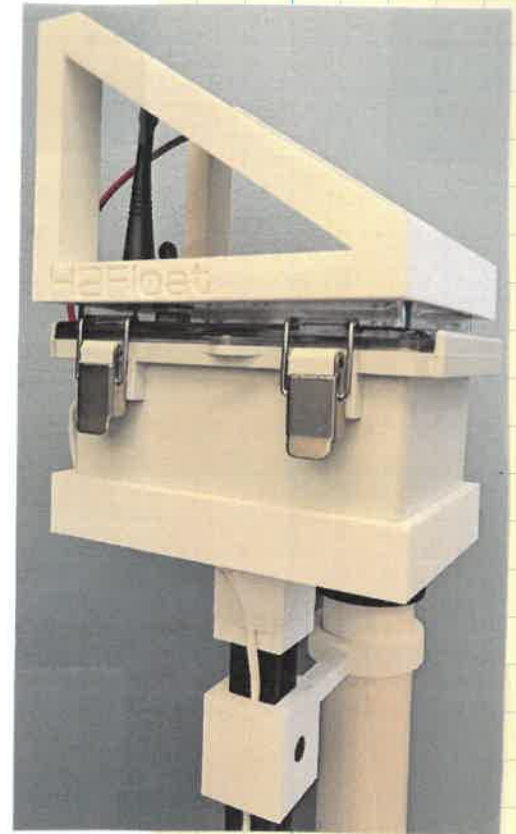
```

#include <SPI.h>           // Include libraries
#include <SD.h>
#include <LoRa.h>
#include <RTCZero.h>
File H2FloatFile;         //Naming the depth data file
RTCZero rtc;              //Create a name (instance) for the RTC stuff
const int CS = 11;        //Chip Select
float Depth;
float Battery;
const byte seconds = 10;   //This will set the current clock time
const byte minutes = 59;
const byte hours = 14;

void setup()
{
  Serial.begin(9600);
  pinMode(CS, OUTPUT);     //Chip select pin (controls SD SPI)
  rtc.begin();             //initialize RTC 24 hour format
  rtc.setTime(hours, minutes, seconds); //Sets the starting clock time
  while(!Serial);          //Must open Serial monitor
  Serial.println("LoRa Receiver");
  if(!LoRa.begin(915E6))   //915 MHz
  {
    Serial.println("LoRa failed");
    while(1);              //If LoRa fails, it jams up
  }
  if(!SD.begin(CS))        //If it does not begin sit in the while loop
  {
    Serial.println("SD Failed");
    while(1);              //Print error command indicate stuck in loop
  }
  bool fileExists = SD.exists("H2Float.txt"); //if the file exists this will be true
  if(fileExists > 0)        //delete existing file and all data
  {
    SD.remove("H2Float.txt");
  }
  H2FloatFile = SD.open("H2Float.txt", FILE_WRITE); //Open file
  H2FloatFile.println("H2Float File");             //At very top of the new file
  H2FloatFile.println("_____");
  H2FloatFile.println("TimeStamp   Depth   Battery Voltage"); //Column headers
  H2FloatFile.close();   //Close file
}

void loop()
{
  int h = rtc.getHours();
  int m = rtc.getMinutes();
  int s = rtc.getSeconds();
  float packetSize = LoRa.parsePacket(); //Get the packet
  while(LoRa.available())
  {

```




```

if(packetSize)
{
  char rcvd = LoRa.read();
  if(rcvd == 'H')                                //Packet identifier
  {
    char rcvd = LoRa.read();
    if(rcvd == 'D')                              //Depth Data Header
    {
      Depth = LoRa.parseFloat();
    }
    else if(rcvd == 'B')                          //Battery Voltage Data Header
    {
      Battery = LoRa.parseFloat();
    }
  }
  H2FloatFile = SD.open("H2Float.txt", FILE_WRITE); //Open and Name file
  if (h < 10)
  {
    H2FloatFile.print("0");
  }
  H2FloatFile.print(h);
  H2FloatFile.print(":");
  if (m < 10)
  {
    H2FloatFile.print("0");
  }
  H2FloatFile.print(m);
  H2FloatFile.print(":");
  if (s < 10)
  {
    H2FloatFile.print("0");
  }
  H2FloatFile.print(s);
  H2FloatFile.print("\t");
  H2FloatFile.print(Depth);                      //Put Depth data in file
  H2FloatFile.print("\t\t");
  H2FloatFile.println(Battery);                   //Put Battery Voltage data in file
  H2FloatFile.close();                           //Close file
} } } }

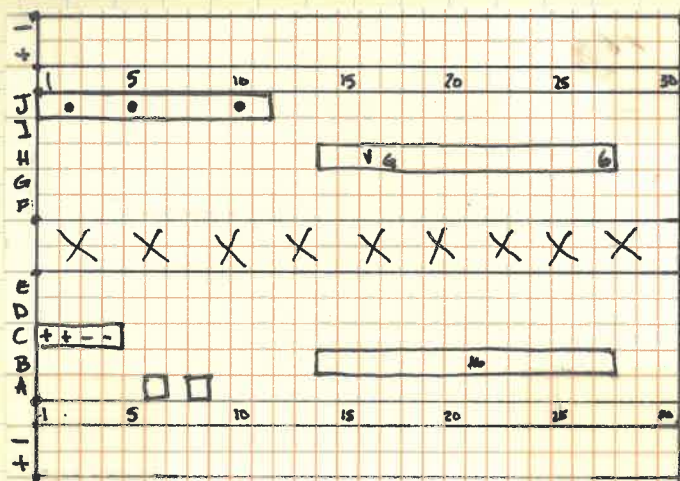
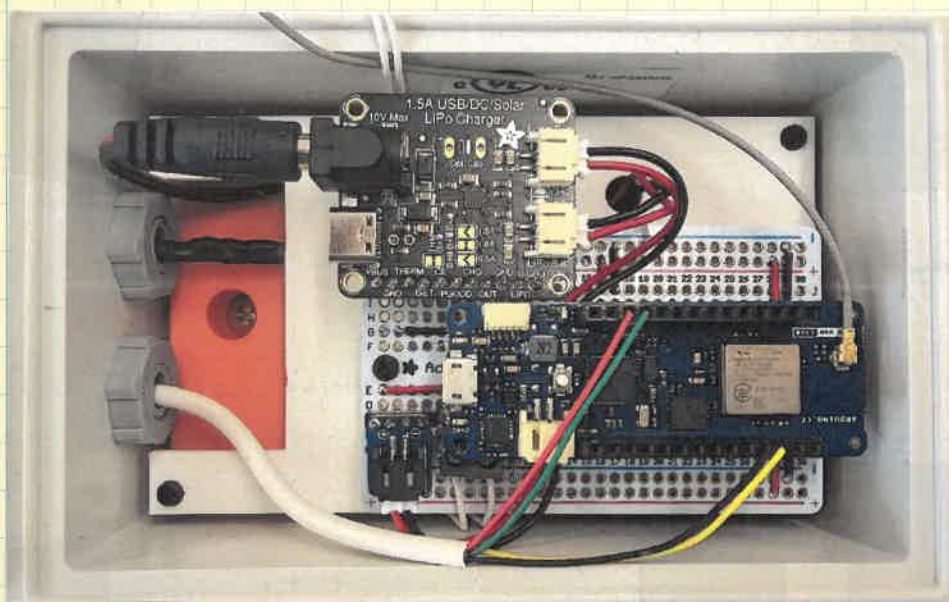
```

//Print SD data To Serial Monitor

```

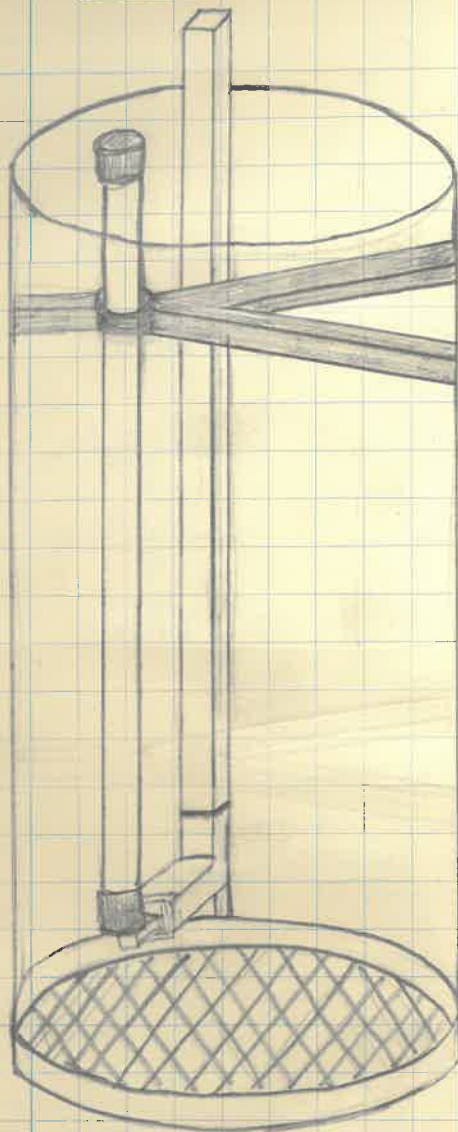
if (h < 10)
{
  Serial.print("0");
}
Serial.print(h);
Serial.print(":");
if (m < 10)
{
  Serial.print("0");
}
Serial.print(m);
Serial.print(":");
if (s < 10)
{
  Serial.print("0");
}
Serial.print(s);
Serial.print("\t");
Serial.print(Depth);
Serial.print("\t\t");
Serial.println(Battery);

```



Summer 2025

Production Model Development



Parts List

- 8" Corrugated Perforated Pipe
- Extruded aluminum Rod
- 1.5" PVC Pipe

3-D Printed Parts

(PETG Carbon Fiber)

- Top PVC Cap (or Store Bought PVC Cap)
- Bottom PVC Cap
- Load cell Support Elbow
- Bottom Grate
- Pipe Supports (2 to 4)

Electronics

- SEN-13879 HX-711
- Load Cell (1kg, 5kg, 10kg)
- MicroSD card Breakout
- MKR WAN 1310 (microcontroller)
- Antenna
- LiPo 6600mAh Battery Backup
- 22/4 Alarm Wire
- Power Switch
- LED Indicator Light

5/19/2025 Load Cell Force Calculations

Loadcell	Pipe Lengths (d)		Volume of Pipe (V_{pipe})	Maximum F_b	
			$\pi \cdot (24.5 \text{ mm})^2 \cdot d$		
1 kg	1 ft	304.8 mm	574773.9 mm ³	6.7093 N	684 g
5 kg	2 ft	609.6 mm	1149547.7 mm ³	12.3461 N	1259 g
	4 ft	1219.2 mm	2299095.4 mm ³	23.6197 N	2408 g
	6 ft	1828.8 mm	3448643.1 mm ³	34.8933 N	3558 g
	8 ft	2438.4 mm	4598190.8 mm ³	46.1669 N	4708 g
10 kg	10 ft	3048 mm	5747738.5 mm ³	57.4406 N	5857 g

$$F_b = -\rho g V$$

$$\rho = 1 \times 10^{-6} \text{ kg/mm}^3$$

$$g = 9807 \text{ mm/s}^2$$

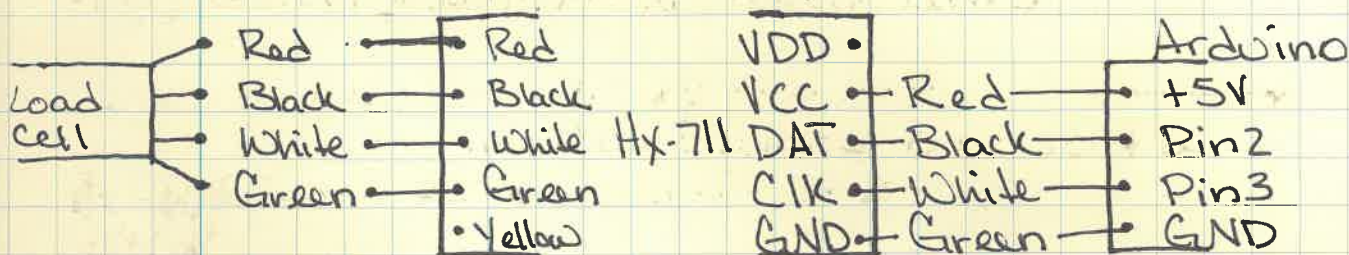
$$V = \pi r^2 \cdot d \text{ (mm)}^3$$

$$\begin{aligned} \text{Volume of Cap } (V_{\text{cap}}) \\ \pi (29.5 \text{ mm})^2 \cdot 40 \text{ mm} \\ = 109358.8 \text{ mm}^3 \end{aligned}$$

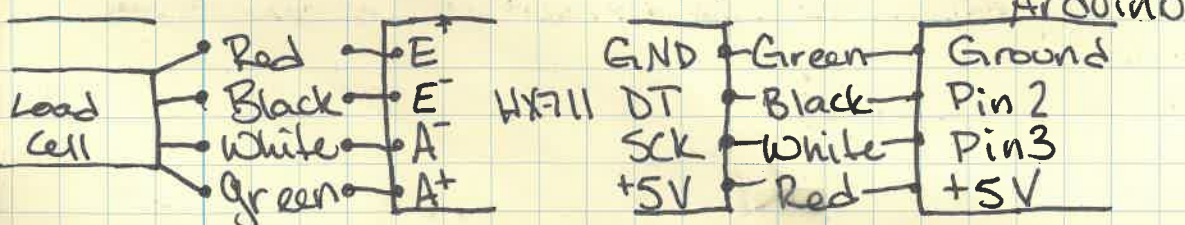
Total Volume is

$V_{\text{pipe}} + V_{\text{cap}} = V$ in F_b calculation

New HX-711 $\Rightarrow$ (SEN-13879) (Red / Bigger)



Old HX-711 (XFW) (green / small)



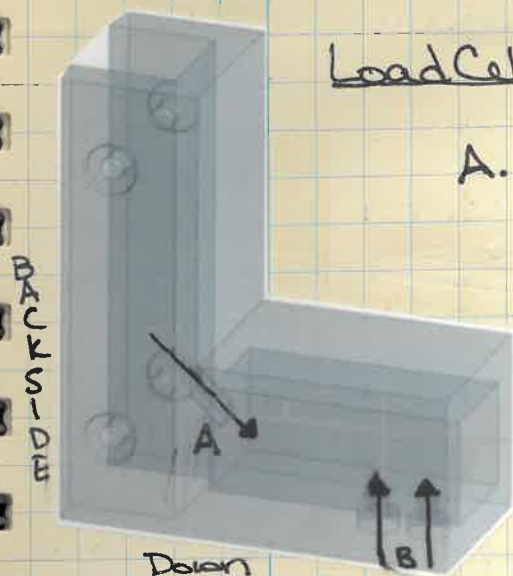
Works With
all Cat
Designs

$\hookrightarrow$ Alarm Wire 22/4

5/20/2025

Assembly of Device Part-By-Part

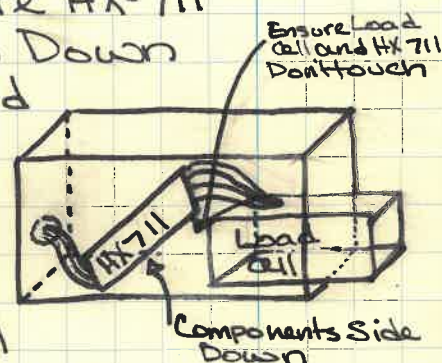
Load Cell Support Elbow HX711 and Load Cell



A. Run 22/4 Alarm Wire through

- Strip Inner 4 wires to attach to HX711 as previously indicated ~~p.40~~ (p.40)
- Solder Inner Wires to HX-711
- Solder 4 Wires from Load Cell to other Side of HX-711 (p.40)
- Pull Alarm Wire Back through toward Back Side of Support Elbow

→ Use Caution to ensure HX-711 Component side faces Down and angles upward without touching Load Cell



B. Attach Load cell wire side in

- ← Using 2 M4 - 0.7 x 20mm Machine Screws/Stainless Steel Pan Phillips Head.

→ Screw both screws into Support Elbow to where they just barely Break the inner plane. Position Load cell over them and ~~screw~~ flush with inner plane then screw them in switching between them as needed to ensure the Load Cell Remains flush with the elbow Support.



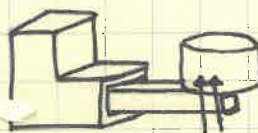
These Components (Load Cell and HX-711) will need to be water proofed using a potting Solution.

- Tape the wire inside the Back of the Support Elbow so it can Lay Flat on a table on the Backside
- Use Some Hot glue to Seal up the wire Exit out the Back ~~and~~
- Pour Potting Mix in to Components Opening and fill to top
- Allow to Fully Cure

5/21/2025

Assembly Continued (Part by Part)Bottom Cap for Floating Pipe to Load Cell

Place Cap on top of Load Cell with Holes lined up. Using 2 x M4-0.7 x 30mm Screws attach Cap to Load cell so cap and Load cell Remain flush.

**EVERBILT™**

750 448

MACHINE SCREWS

M4-0.7 mm x 30 mm

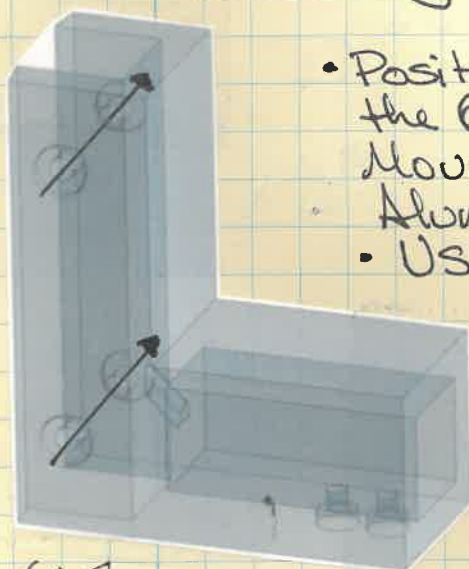
PAN HEAD PHILLIPS

STAINLESS

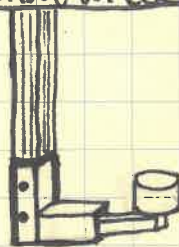
2 PK



MADE IN TAIWAN

Attaching Full Elbow Assembly to Extruded Aluminum Rod

- Position Elbow flush with the Bottom of the Aluminum Rod. Mark the Position of the Mounting Holes of the Elbow on the Aluminum Rod.
- Use a 3/16" Drill Bit to Make Holes through the Center of the Aluminum Rod.
- Use 2 x #10-24 x 1.5" screws and 2 x #10-24 Lock nuts to firmly attach Elbow to Rod



WEATHERMAX™ 126951

Stainless Steel

Pan Phillips Machine

Máquina Cilíndrica Phillips
Acero Inoxidable

10-24 x 1-1/2 Pcs 8

WEATHERMAX™ 126896

Stainless Steel

Nylon Insert Lock Nuts

Contratuercas Inserto
de Nílon
Acero Inoxidable

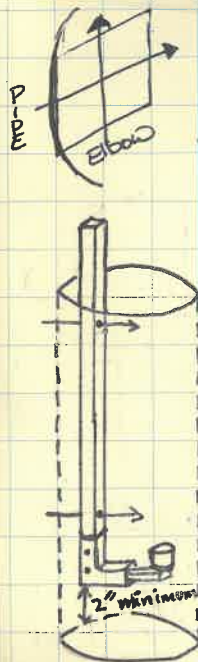
10-24

Pcs 8

Attach Aluminum Rod to Corrugated Pipe

★ This Requires the Same #10-24 screw and Nut as the Previous Step.

↳ Additionally this Requires the Use of a Sufficiently Sized Washer.

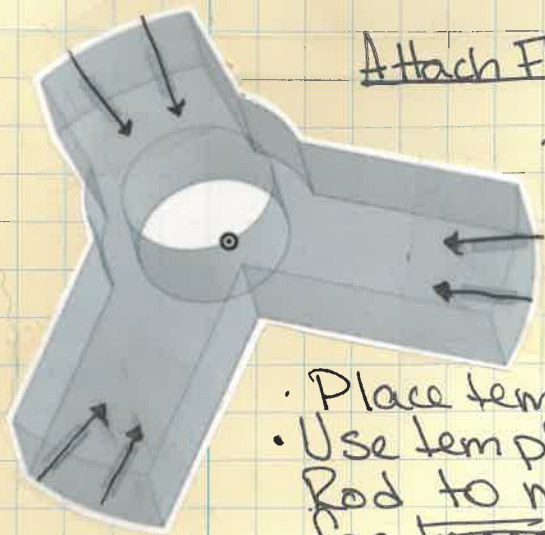


- These drill Holes are made in the crossed direction of the drill Holes for the Elbow.
- Position the Aluminum Rod with Elbow assembly at the Bottom of the Corrugated Pipe with at Least 2" Clearance from the Very Bottom
- Drill Holes(2) at Top and Bottom of Aluminum Rod and through Corrugated pipe to attach Rod to Pipe (make sure Holes line up with smaller diameter corrugation). Attach Firmly and Vertically Level. (Utilized Green line on Out Side of Pipe)



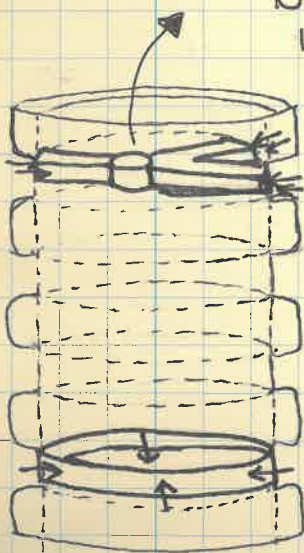
Assembly Continued (Part by Part)

5/27/2025

Attach Floating Pipe Support to Inside of Corrugated Pipe

This Required the Creation of a Template to ensure mounting holes would line up with drilled Holes in Corrugated pipe.
(See page 44 for template)

- Place template on Smaller diameter Corrugation
- Use template Lined up on Center of Aluminum Rod to mark then drill Hole positions for ~~inner~~ the Floating Pipe Support.
 - Drill Holes Using $3/16$ " drill bit
- Position and attach Pipe Support Using $6 \times \#10-24 \times 3/4$ " screws →
 - ↳ Support Should Be mounted about 7-9" below top of corrugated pipe

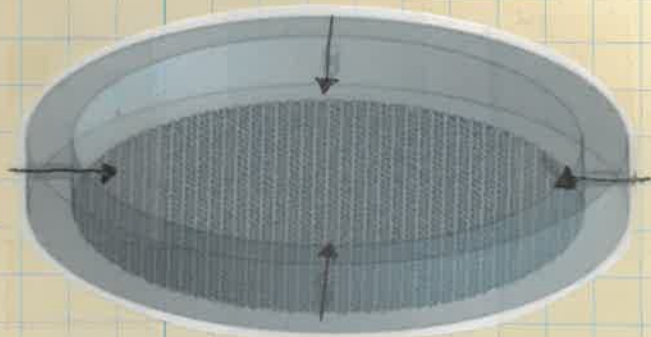


(Not to Scale)
Positions Inexact

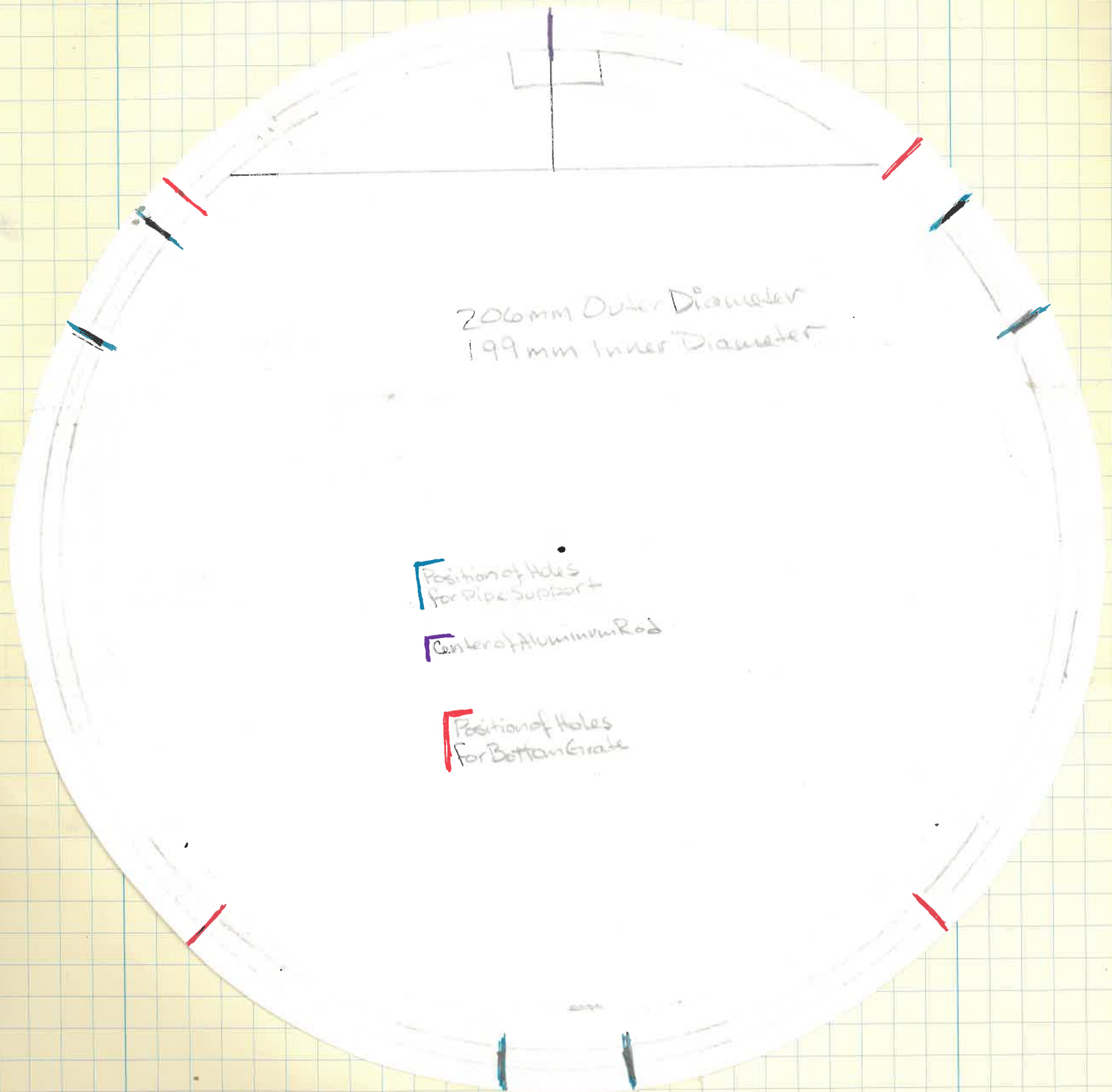
Attach Bottom Grate to Inside of Corrugated Pipe

This also Requires Use of a Template.
(See Page 44 for Template)

- Using the Template around the Bottom most Smaller diameter Corrugation and a $3/16$ " drill bit make holes in the Corrugated pipe
- Insert Bottom Grate and line up holes
- Attach Using $\#10-24 \times 3/4$ " screws



5/22/2025 Template Design for Positioning Drill Holes
Required for Inner Mounted Grate and Pipe Support

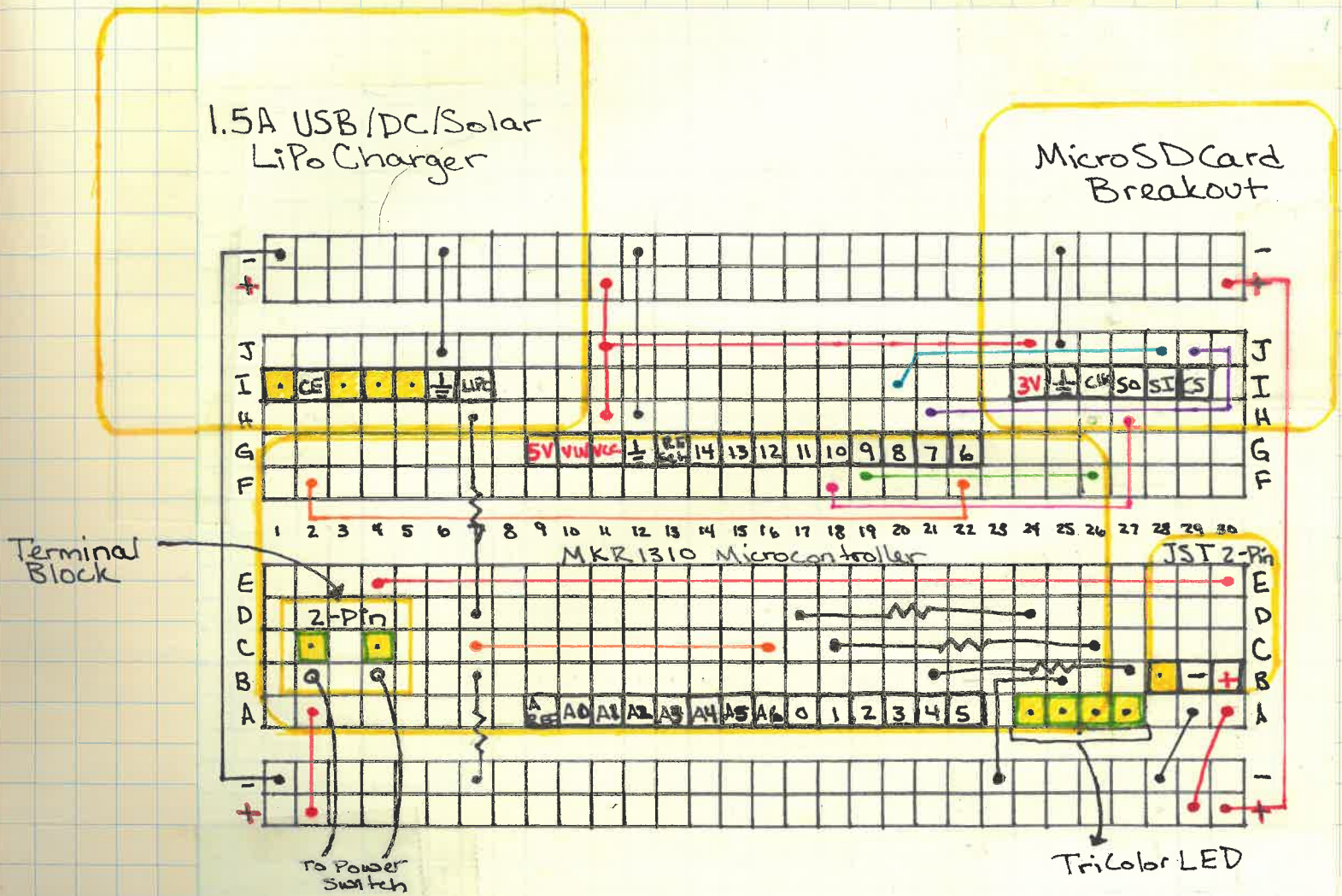


- Affix String to the Outer Circle and mark the positions of the holes with the color code
- To Use Wrap around smaller diameter of Corrugated pipe in desired location and line up the Center of the metal Rod mark on the metal Rod.

Assembly Continued (Electronics)

5/29/2025

Circuit Board Design and Components Connections (Arduino MKR1310 specific)



Board Components (Soldered)

(Electronics Attached Via Header Pins)

MKR1310 Microcontroller

JST 2-Pin Breakout Connector

1.5A USB/DC/Solar LiPo Charger

MicroSD Card Reader Breakout

2-Pin Terminal Block (Power Switch)

2 Resistors 200k Ω

Tri Color LED

Wire (22/4) From Load Cell to MKR1310 (Top Side Connection)

+5V Red $\rightarrow$ VCCBlack $\rightarrow$ Pin 2White $\rightarrow$ Pin 3Green $\rightarrow$ GND $\perp$

• Soldered Pin (Unused)

◻ Indirect Connect Soldered Pin

◻ Soldered Wire Connection
(Direct Labeled Pins)

◻ Resistor

All MKR1310 Top Side Connections
are Header Pin soldered at each
Labeled Position